

Q1. DIRECTIONS *for question 1:* Type in your answer in the input box provided below the question.

X, Y and Z are positive numbers. $[N]$ denotes the integral part of N and $\{N\}$ denotes the fractional part of N. If the following equations are given,

$$[X] + \{Y\} + [Z] = 11.2$$

$$[Y] + \{Z\} + [X] = 10.3$$

$$[Z] + \{X\} + [Y] = 9.4$$

find the value of $[X + Y - Z]$.

Q2. DIRECTIONS *for questions 2 and 3:* Select the correct alternative from the given choices.

Among 300 students in a coaching institute, 180 took the CAT exam, 150 took the XAT exam and at least 50 students took neither the CAT nor the XAT. If the number of students who took both the exams is at least N and at most M, find the value of $M + N$.

☐ a) 190

☐ b) 210

☐ c) 180

☐ d) 230

Q3. DIRECTIONS *for questions 2 and 3:* Select the correct alternative from the given choices.
If the LCM of P and 180 is 720, and the HCF of P and 150 is 30, find the sum of the digits of P .

- ☐ a) 9
- ☐ b) 3
- ☐ c) 6
- ☐ d) Cannot be determined

Q4. DIRECTIONS *for questions 4 and 5:* Type in your answer in the input box provided below the question.

Eight identical balls are to be placed in three different boxes. If each box can hold any number of balls, then in how many ways can the balls be placed in the boxes so that no box remains empty?

Q5. DIRECTIONS *for questions 4 and 5:* Type in your answer in the input box provided below the question.

What is the remainder when the number 382382 up to 300 digits is divided by 101?

Q6. DIRECTIONS *for question 6:* Select the correct alternative from the given choices.

Let P, Q, S, R, T, U and V represent the seven distinct digits from 0 to 6, not necessarily in that order. If PQ and RS are both two-digit numbers adding up to the three-digit number TUV, find the value of V.

- ☐ a) 3
- ☐ b) 6
- ☐ c) 5
- ☐ d) **Cannot be determined**

Q7. DIRECTIONS *for question 7 and 8:* Type in your answer in the input box provided below the question.

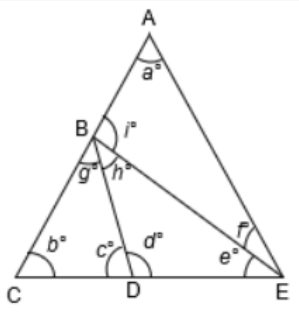
Sethi and Wilson play a snooker match comprising nine games. The winner is decided by the method of 'Race to 5', i.e., the first person to win five games is declared the winner and the match ends the moment the winner is declared. In how many ways (i.e., combinations of number of games won by each player) can the match conclude?

Q8. DIRECTIONS *for question 7 and 8:* Type in your answer in the input box provided below the question.

Let a , b , c and d be positive integers satisfying the equations $\log_a b = \frac{5}{4}$, $\log_c d = \frac{3}{2}$ and $a - c = 32$. Find $d - b$.

Q9. DIRECTIONS for questions 9 and 10: Select the correct alternative from the given choices.

The figure below gives certain information and it is followed by two quantities, I and II. Compare the two quantities and then select the answer option which is true regarding the relationship between them.



I. $g + c - e$

II. $a + f$

- ☐ a) $I > II$
- ☐ b) $II > I$
- ☐ c) $I = II$
- ☐ d) The relationship cannot be determined from the given information.

Q10. DIRECTIONS *for questions 9 and 10:* Select the correct alternative from the given choices.

If the profit obtained from selling five apples is the same as the selling price of three apples, what is the profit percentage from selling one apple?

- ☐ a) 75%
- ☐ b) 100%
- ☒ c) 150% ✓ **Your answer is correct**
- ☐ d) 175%

Q11. DIRECTIONS *for question 11:* Type in your answer in the input box provided below the question.

A triathlon is an athletic event in which each participant has to perform three activities, swimming, running and cycling. Shiva wanted to participate in a triathlon. He prepared himself for it by going for workouts every morning and evening. The ratio of his speeds of cycling, running and swimming is $10 : 4 : 1$ respectively. One day, in the morning workout, he covered equal distances while performing each activity and in the evening workout, on the same day, he covered a total distance which was five times that he covered in the morning workout, spending equal times on each of the three activities. If the total time he spent on his workouts that day was 348 minutes, find the total time (in minutes) he spent on cycling that day.

Q12. DIRECTIONS *for questions 12 to 16:* Select the correct alternative from the given choices.

Three positive real numbers p , q and r satisfy the equations $q - p = r - q$ and $pqr = 6$. Find the minimum possible value of q .

☐ a) 2

☐ b) 1

☐ c) $\frac{1}{6^3}$

☐ d) $\frac{2}{6^3}$

Q13. DIRECTIONS for questions 12 to 16: Select the correct alternative from the given choices.

Given that $3m = 5n$ and $p = 3q$, where m, n, p and q are all real numbers, and $A = pqm^2 + 3q^2n^2$ and $B = p^2mn + 3mpnq$, find the value of $\left| \frac{A+B}{A-B} \right|$

- ☐ a) $\frac{31}{14}$
- ☐ b) $\frac{14}{25}$
- ☐ c) $\frac{25}{31}$
- ☐ d) **Cannot be determined**

Q14. DIRECTIONS *for questions 12 to 16:* Select the correct alternative from the given choices.

If the equation $cx^2 + ax + b = 0$ has two positive roots, K and $3K$, and $3(b - a) = 5c$, then find $K^{\frac{1}{K}}$.

- ☐ a) $\frac{1}{\sqrt[3]{3}}$
- ☐ b) $\frac{1}{27}$
- ☐ c) $3^{1/3}$
- ☐ d) 1

Q15. DIRECTIONS *for questions 12 to 16:* Select the correct alternative from the given choices.

When p is a prime number and n is any natural number not divisible by p , then $(n^p - 1) - 1$ is divisible by p . Find the remainder when 32^{1000} is divided by 97.

☐ a) 93

☐ b) 31

☐ c) 33

☐ d) 62

Q16. DIRECTIONS *for questions 12 to 16:* Select the correct alternative from the given choices.

A dishonest sugar merchant, while purchasing goods from the dealer, manages to receive 20% more sugar than the quantity of sugar he is supposed to receive. He later sells the sugar to his customers in such a way that for any customer, 30% of the sugar actually delivered must be added to get the original quantity of sugar asked for by the customer. The merchant sells the sugar to his customers at cost price (i.e., the price demanded by the dealer), but then charges them an additional amount claiming it as his transportation cost. If the additional amount that the merchant charges equals 10% of the cost that he incurred on the sugar that he is actually delivering to them, find his net gain percent. (Assume that the merchant's actual transportation cost is zero)

- ☐ a) **66%**
- ☐ b) **60%**
- ☐ c) **68%**
- ☐ d) **71.6%**

Q17. DIRECTIONS *for question 17:* Type in your answer in the input box provided below the question.

If $\log_5 3$, $\log_5 2^x$ and $\log_5 (3^x + 13/3)$ are in arithmetic progression, find x .

Q18. DIRECTIONS *for questions 18 and 19:* Select the correct alternative from the given choices.

A solution comprises two chemicals X and Y. The price of X is ₹150 per litre and that of Y is ₹300 per litre. The maximum that can be spent for making the solution is ₹2,100. If X and Y weigh 12 kg/litre and 15 kg/litre respectively, and the solution must contain each of the chemicals to the extent of at least 20% by weight, then the maximum weight of the solution that can be made, is closest to

- ☐ a) 140 kg.
- ☐ b) 130 kg.
- ☐ c) 150 kg.
- ☐ d) 160 kg.

Q19. DIRECTIONS *for questions 18 and 19:* Select the correct alternative from the given choices.

If $y = |x + 1| - |x - 2|$, then which of the following is true of y ?

☐ a) $-3 \leq y \leq 0$

☐ b) $-3 \leq y \leq 3$

☐ c) $y \leq -3$

☐ d) $y \geq 3$

Q20. DIRECTIONS *for question 20:* Type in your answer in the input box provided below the question.

If x , y and z are positive real numbers, find the ratio $\frac{y}{z}$ for which the value of $\left(\frac{x}{y} + \frac{z}{12x} + \frac{4y}{x} + \frac{x}{3z}\right)$ is the minimum.

Q21. DIRECTIONS *for question 21:* Select the correct alternative from the given choices.

Amar borrowed ₹15,000 each from two banks, RIB and SIB. RIB charged a 20% rate of interest, compounded annually, while SIB charged an equal rate of interest but compounded semi-annually. If Amar repaid the both the loans in full, each in a single payment, two years after he borrowed the money, what is the ratio of the amount repaid by Amar to RIB to that repaid to SIB?

- ☐ a) **14400 : 14641**
- ☐ b) **100 : 121**
- ☐ c) **12100 : 14400**
- ☐ d) **16000 : 17600**

Q22. DIRECTIONS *for question 22:* Type in your answer in the input box provided below the question.
In the number 160^{4320} , after how many digits from the right will you encounter the first odd digit?